Visual Place Recognition with Geometric Re-ranking: A Detailed Study of the Accuracy–Compute Trade-off and Adaptive Re-ranking

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Abstract

Visual Place Recognition (VPR) localizes a query image by retrieving geo-tagged database images depicting the same physical place. A standard modern pipeline combines fast global-descriptor retrieval with optional local geometric verification, which re-ranks the top candidates using feature matching and robust model estimation (RANSAC) through an inlier-count scoring function. While geometric re-ranking often improves top-1 localization under perceptual aliasing and domain shift, it adds significant computational cost. This report provides an in-depth description of the core algorithms underlying descriptor-based VPR, including the architectural roles of backbones and aggregation, the metric-learning principles used to train retrieval embeddings, and the mechanisms of learned matchers such as SuperGlue, SuperPoint+LightGlue, and LoFTR. We then present extensive experimental results comparing multiple retrieval models (CosPlace, NetVLAD, MixVPR, MegaLoc) and multiple matchers (LoFTR, SuperPoint+LightGlue, SuperGlue) across SF_XS, Tokyo_XS, and cross-domain SVOX (Sun/Night) subsets, reporting Recall@N and runtime. We also validate that dot product and ℓ_2 distance retrieval yield identical results in our setting due to descriptor normalization. Finally, we present an adaptive re-ranking extension that aims to preserve re-ranking gains while reducing average runtime by triggering costly verification only for hard queries. The adaptive policy uses a single top-1 geometric "probe" (inlier count) and a logistic regression gate to decide whether to run full top-K matching, yielding a tunable accuracy–compute trade-off.

The code used to reproduce the experiments and results presented in this report is available at: https://github.com/dariush-vaghei/AML_Visual_Place_Recognition.

1. Introduction

Visual Place Recognition (VPR) asks a deceptively simple question: given a query image, where was it taken? Formally, the input is a single RGB image, and the output is a geographic hypothesis obtained by matching it against a database of reference images with known locations. In robotic navigation, VPR is used for re-localization and loop closure; in autonomous driving it can provide map-level anchoring; in augmented reality it enables persistent content aligned to the real world. Despite its apparent similarity to generic image retrieval, VPR has distinct challenges. Images of the same place can vary drastically across time and sensing conditions: illumination, weather, seasons, occlusions by dynamic objects, motion blur, and viewpoint changes that may reduce overlap between query and reference.

A central difficulty is *perceptual aliasing*: different places can look globally similar (e.g., building facades, road intersections, or corridors), causing a global descriptor to retrieve an incorrect but visually similar location. Conversely, the same place may appear dissimilar under strong viewpoint or illumination shifts, making retrieval brittle. Practical VPR systems therefore combine two complementary signals: a *global* descriptor embedding for fast nearest-neighbor retrieval and a shortlist of candidates, and a *local geometric* verification stage that matches features and enforces consistency to identify the true physical match.

This design introduces a fundamental trade-off. Global retrieval is efficient, typically taking milliseconds per query when descriptors are precomputed and indexed. Geometric verification is far more expensive because it requires running a matcher and RANSAC for each of the top-K candidates. The motivating question of this work is therefore not only "does re-ranking improve Recall@1?", but rather: *how should we spend compute to obtain the best accuracy for a given budget?*

Following the project specification, we first analyze the standard pipeline and quantify how retrieval and re-ranking behave across datasets and conditions. We then focus on an extension: *adaptive re-ranking*, based on the observation that many queries are easy and do not require full verification. If the top-1 retrieval is reliable, we skip re-ranking; otherwise, we spend compute to refine the result. In this report we provide (i) a theoretical grounding of the models and algorithms used and (ii) an experimental analysis of the adaptive workflow, including compute-aware evaluation of the accuracy–cost trade-off.

2. Related Work

VPR has been studied extensively, spanning classical handcrafted representations and modern deep embedding methods. Here we focus on work directly relevant to the project pipeline: learned global descriptors for retrieval, learned local matching for verification, and efficiency-driven strategies that selectively apply expensive computation.

Global descriptors and aggregation. Deep VPR methods typically combine a backbone feature extractor with a retrieval-oriented aggregation layer. NetVLAD introduced a differentiable VLAD-like residual aggregation mechanism, which became a strong retrieval baseline. Later approaches improved scaling and robustness through better training objectives and more compact embeddings, e.g., CosPlace and other cosine-embedding approaches. More recent work such as MixVPR emphasizes improved feature aggregation by mixing spatial information before projection, which often improves robustness to viewpoint changes. MegaLoc represents a strong retrieval system designed to generalize across domains and conditions, often yielding high Recall@1 with no refinement.

Metric learning and mining. Most deep retrieval models are trained with metric learning: triplet losses, contrastive objectives (InfoNCE), and variants that depend heavily on hard negative selection. Negative mining is particularly important in VPR because perceptual aliasing produces hard negatives that are visually close in embedding space.

Local matching and geometric verification. Geometric verification is a classical tool in structure-from-motion and image matching. RANSAC provides robust estimation under outliers, enabling inlier counting as a verification score. In learned matching, SuperPoint introduced a self-supervised keypoint detector/descriptor. SuperGlue introduced a learned matcher with global reasoning, significantly improving correspondence quality in ambiguous scenes. LightGlue improved efficiency while maintaining strong performance. LoFTR proposed detector-free matching using transformer-style attention over dense features, often robust under difficult texture conditions.

Efficiency and adaptive pipelines. A recurring theme in retrieval systems is conditional computation. When retrieval confidence is high, verification may not be needed. Our extension formalizes this idea by using the number of inliers between the query and its top-1 retrieval as a confidence feature and learning a lightweight gate. This connects to broader work on cascaded retrieval, calibrated confidence, and compute-aware inference.

3. Methodology

3.1. Global descriptor retrieval

3.1.1. Problem formulation

Let f_θ map an image I to a descriptor $z = f_\theta(I) \in \mathbb{R}^D$. Given a query image q and a database $\{I_i\}_{i=1}^N$, VPR retrieval ranks candidates by similarity:

$$\mathrm{TopK}(q) = \arg\max_i \ \mathrm{sim}(f_\theta(q), f_\theta(I_i)), \tag{1}$$

returning the top-K nearest neighbors. In this project we use $K = 20$ to support later re-ranking.

The effectiveness of retrieval depends on whether the embedding space organizes images such that same-place images cluster and different places separate. This is not guaranteed by generic classification training; it is explicitly enforced by metric learning.

3.1.2. Backbone networks and ResNet-50

Few models in this project rely on a CNN backbone, typically ResNet-50, which outputs a spatial feature map. ResNet-50 is built from residual blocks

$$y = x + F(x), \tag{2}$$

where skip connections stabilize optimization and enable deeper networks. In the context of VPR, the backbone provides multi-level features ranging from low-level texture and geometry to higher-level structural cues. However, the backbone alone does not define a VPR model: the final place descriptor is obtained after an aggregation stage that compresses the spatial feature map into a compact vector. This aggregation step is where VPR methods primarily differ.

3.1.3. Aggregation mechanisms and model explanations

(1) NetVLAD: residual aggregation over learned prototypes. NetVLAD takes local descriptors x_i from the backbone feature map and a set of learned cluster centers c_k. Each local descriptor is softly assigned to each center via weights a_{ik} (learned by a small network), and NetVLAD aggregates residuals:

$$v_k = \sum_i a_{ik}(x_i - c_k). \tag{3}$$

The final descriptor concatenates the v_k vectors, often followed by normalization and dimensionality reduction. Intuitively, NetVLAD does not simply average features; it records how local features deviate from prototypes. This captures richer statistics and often improves discrimination under aliasing, but at the cost of larger representations.

(2) CosPlace: compact cosine-embedding retrieval. CosPlace targets scalable retrieval with compact descriptors. While the project uses CosPlace as a black-box model, its

key design goal is to learn a descriptor space where cosine similarity aligns with place identity without requiring overly large embeddings. In practice, compact descriptors reduce memory and accelerate indexing, making CosPlace a good baseline for large-scale systems.

(3) MixVPR: feature mixing for better global descriptors. MixVPR focuses on improving aggregation by mixing spatial information before final pooling/projection. A backbone produces a set of spatial tokens (or local descriptors). Mixing layers (often MLP-like or attention-inspired) combine information across tokens, allowing the model to learn how evidence from different regions should interact. This is especially useful in VPR because different regions can be informative depending on viewpoint and occlusion. Rather than treating all spatial positions independently and then pooling, MixVPR can learn cross-region dependencies.

(4) MegaLoc: strong generalization and high baseline recall. MegaLoc represents a strong retrieval system designed to generalize across conditions and domains. From a system perspective, its role in this project is crucial: it tests whether re-ranking still helps when retrieval is near-saturated. If retrieval is already correct at rank 1 most of the time, re-ranking has limited upside and can sometimes harm performance due to matcher noise.

3.1.4. Similarity metrics: dot product vs. ℓ_2

We compare dot product and Euclidean distance for retrieval ranking and observe identical results. This is expected because VPR descriptors are typically ℓ_2-normalized. For normalized vectors, dot product and ℓ_2 induce the same ranking:

$$\|x - y\|_2^2 = 2 - 2x^\top y \quad \text{when } \|x\|_2 = \|y\|_2 = 1. \tag{4}$$

Thus, this experiment primarily validates that our pipeline uses normalized descriptors correctly.

3.2. How VPR models are trained: metric learning

Although we use pretrained models, understanding training is essential to interpret behavior and failure modes. Training data provides positives (same place) and negatives (different places). Losses enforce that positives are close and negatives are far.

Triplet loss enforces a margin ranking constraint:

$$\mathcal{L}_{\text{triplet}} = \max\left(0,\, d(z_q, z_p) - d(z_q, z_n) + m\right).$$

Contrastive/InfoNCE losses push a positive to be the most similar among a set of candidates:

$$\mathcal{L}_{\text{con}}(q, p) = -\log \frac{\exp(\text{sim}(z_q, z_p)/T)}{\sum_j \exp(\text{sim}(z_q, z_j)/T)}.$$

Hard negative mining is crucial because most negatives are too easy and provide little learning signal.

3.3. Geometric re-ranking

3.3.1. Local matching as a verification signal

Retrieval provides a shortlist of plausible candidates but does not prove geometric consistency. Local matching attempts to establish correspondences between visual elements in the query and candidate. If the two images depict the same place with sufficient overlap, many correspondences should be consistent under a geometric model.

3.3.2. Matcher explanations

SuperPoint: detection and description. SuperPoint is a CNN that outputs a keypoint heatmap and a dense descriptor map. Keypoints are selected by finding peaks in the heatmap, and descriptors are sampled from the descriptor map.

SuperGlue: learned assignment with global reasoning. SuperGlue takes two sets of keypoints and descriptors and performs matching as a learned assignment problem using attention-based message passing. This global reasoning helps resolve ambiguity in repetitive patterns.

SP-LightGlue: efficiency-oriented learned matching. LightGlue follows a similar philosophy (global reasoning over keypoints) but is designed to be lightweight and fast.

LoFTR: detector-free dense matching. LoFTR avoids explicit keypoint detection and matches dense features using transformer-style attention, often robust in low-texture or blurred scenes.

3.3.3. RANSAC and inliers

Tentative correspondences contain outliers. RANSAC robustly fits a geometric model by sampling minimal correspondence sets, estimating a model, and counting consistent matches. Matches that agree with the final model are *inliers*. We use the inlier count as a re-ranking score:

$$s(q, c) = \text{inliers}(q, c).$$

Re-ranking sorts the K candidates by $s(q, c)$.

3.4. Adaptive re-ranking (extension)

3.4.1. Motivation

Always-on geometric re-ranking requires running local matching and RANSAC for all K retrieved candidates, which dominates runtime when $K = 20$. However, many queries are easy: the retrieval top-1 prediction is already correct, and additional verification provides no benefit. Adaptive re-ranking formulates geometric verification as conditional computation, aiming to retain most of the accuracy gains of re-ranking while reducing average cost.

Let t_r denote retrieval time and t_m the cost of one match+RANSAC evaluation. Always-on re-ranking costs

$$t_{\text{full}} = t_r + Kt_m. \tag{5}$$

Adaptive re-ranking always performs a single top-1 probe and only evaluates the remaining $K - 1$ candidates for a fraction p of hard queries:

$$t_{\text{adapt}} = t_r + t_m + p(K - 1)t_m. \tag{6}$$

When p is small, this yields substantial computational savings.

3.4.2. Adaptive policy

Starting from the standard retrieval output (a ranked top-K list), the adaptive policy always performs a cheap probe by matching the query q with the top-1 retrieved candidate c_1 and computing the inlier count

$$I_1(q) = \text{inliers}(q, c_1). \tag{7}$$

A logistic regression maps this scalar confidence feature to the probability that the retrieval top-1 is correct:

$$P(\text{top1 correct} \mid I_1) = \sigma(wI_1 + b). \tag{8}$$

Using a threshold τ, queries are classified as EASY or HARD. EASY queries keep the retrieval ranking, while HARD queries trigger full top-K matching and re-ranking by inlier count.

3.4.3. Training, calibration, and domain handling

The logistic regression is trained on a separate training split using tabular data with one row per query. The single feature is the top-1 inlier count I_1, and the binary label indicates whether the retrieval top-1 prediction is correct under a 25 m localization threshold. Training data are generated by first computing full top-K geometric verification outputs, enabling label construction without bias.

Because inlier-count distributions differ significantly across illumination conditions, two domain-specific gates are trained: LR-Sun and LR-Night. LR-Sun is trained on the SVOX Sun subset, while LR-Night is trained on the SVOX Night subset. Each gate is trained by pooling data across retrieval backbones and matchers within the same domain, capturing the dominant domain shift while avoiding over-specialization.

The threshold τ controls the accuracy–compute trade-off. We sweep τ on a validation set (SF_XS) and select operating points near the knee of the Recall@1–compute curve. This yields $\tau_{\text{Sun}} = 0.15$ and $\tau_{\text{Night}} = 0.25$, reflecting the stricter calibration required under night-time conditions due to lower inlier counts and higher ambiguity.

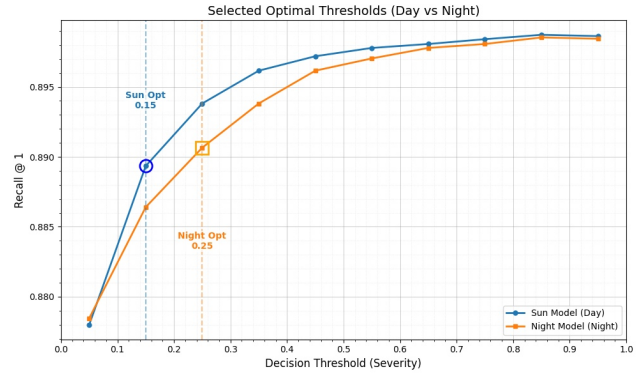


Figure 1. Decision-threshold sweep for the logistic-regression gate. Adaptive Recall@1 increases rapidly at low thresholds and then saturates. We select $\tau_{\text{Sun}} = 0.15$ and $\tau_{\text{Night}} = 0.25$ as near-knee operating points that capture most of the gain while limiting additional compute.

3.4.4. Runtime-aware evaluation

Adaptive evaluation is performed end-to-end. For each query, the policy always evaluates the top-1 match and evaluates the remaining top-$(K - 1)$ candidates only for HARD queries. We report adaptive Recall@$\{1, 5, 10, 20\}$ together with compute proxy metrics: hard-query rate and savings relative to always matching all K candidates. Savings is computed as the fraction of avoided match evaluations:

$$\text{savings} = 1 - \frac{\text{match evals}}{K}. \tag{9}$$

These metrics directly quantify the accuracy–compute trade-off enabled by adaptive re-ranking.

4. Experiments

4.1. Datasets and protocol

We follow the project-provided subsets: SF_XS, Tokyo_XS, and SVOX for cross-domain evaluation under Sun and Night conditions. We use Recall@N under a tolerance $\tau = 25$m and retrieval shortlist size $K = 20$. For re-ranking, images are resized to 512×512 for local matching.

The experimental workflow is staged: (i) retrieval-only baselines, (ii) always-on re-ranking (upper bound on geometric gains), (iii) adaptive re-ranking (conditional computation).

Runtime reporting note. Our runtime measurements were collected in a distributed setup: team members executed experiments on their own environments, including local machines and Google Colab instances (free and paid tiers with different GPU models). As a result, the absolute wall-clock times can exhibit variability across runs due to hardware heterogeneity, background system load, and runtime configuration differences. We therefore interpret runtime results as indicative trends rather than strictly comparable absolute values.

Global	Local	SF_XS R@1/R@5/R@10/R@20	TOKYO_XS R@1/R@5/R@10/R@20	SVOX (Sun) R@1/R@5/R@10/R@20	SVOX (Night) R@1/R@5/R@10/R@20
CosPlace	None	63.1 / 74.8 / 78.6 / 81.3	65.1 / 79.7 / 86.0 / 89.5	62.3 / 78.5 / 84.5 / 88.8	33.3 / 51.4 / 59.7 / 67.6
	LoFTR	77.4 / 79.8 / 80.5 / 81.3	84.8 / 87.9 / 88.6 / 89.5	83.2 / 87.1 / 87.8 / 88.8	61.2 / 65.1 / 66.2 / 67.6
	SP-LightGlue	77.6 / 80.1 / 80.6 / 81.3	82.9 / 86.7 / 87.9 / 89.5	83.4 / 87.0 / 87.9 / 88.8	60.9 / 64.9 / 66.3 / 67.6
	SuperGlue	76.7 / 79.4 / 80.5 / 81.3	82.5 / 86.3 / 88.6 / 89.5	81.3 / 85.9 / 87.4 / 88.8	59.4 / 64.5 / 65.9 / 67.6
MixVPR	None	69.6 / 78.7 / 81.1 / 83.6	76.5 / 88.6 / 92.1 / 94.3	89.4 / 92.7 / 94.6 / 95.4	62.6 / 80.2 / 83.7 / 87.4
	LoFTR	79.4 / 82.8 / 83.5 / 83.6	90.5 / 92.7 / 94.0 / 94.3	92.9 / 94.6 / 95.3 / 95.4	82.0 / 85.9 / 86.9 / 87.4
	SP-LightGlue	80.7 / 82.6 / 83.1 / 83.6	89.2 / 92.4 / 94.0 / 94.3	92.5 / 94.4 / 95.2 / 95.4	81.3 / 85.5 / 86.6 / 87.4
	SuperGlue	79.3 / 82.2 / 83.2 / 83.6	87.0 / 92.4 / 93.3 / 94.3	89.6 / 93.6 / 95.0 / 95.4	81.0 / 85.7 / 86.9 / 87.4
NetVLAD	None	42.2 / 53.7 / 60.8 / 65.3	54.0 / 64.4 / 69.5 / 74.6	37.1 / 54.6 / 62.2 / 69.0	8.5 / 18.1 / 22.7 / 29.4
	LoFTR	61.5 / 63.9 / 64.6 / 65.3	68.9 / 71.7 / 73.0 / 74.6	64.6 / 66.5 / 67.4 / 69.0	25.2 / 27.5 / 28.8 / 29.4
	SP-LightGlue	62.3 / 64.5 / 64.9 / 65.3	68.9 / 70.8 / 73.0 / 74.6	65.1 / 66.7 / 67.8 / 69.0	27.7 / 27.2 / 27.9 / 29.4
	SuperGlue	61.1 / 64.4 / 64.9 / 65.3	67.0 / 71.1 / 72.7 / 74.6	64.3 / 66.3 / 67.4 / 69.0	24.8 / 27.1 / 28.4 / 29.4
MegaLoc	None	86.9 / 90.4 / 91.2 / 91.5	**95.6 / 97.8 / 98.7 / 99.0**	**97.2 / 99.3 / 99.5 / 99.6**	**96.5 / 98.7 / 99.0 / 99.3**
	LoFTR	86.5 / 89.5 / 90.7 / 91.5	94.3 / 97.8 / 98.7 / 99.0	97.2 / 99.2 / 99.5 / 99.6	92.6 / 98.5 / 99.0 / 99.3
	SP-LightGlue	**87.0 / 90.7 / 91.2 / 91.5**	94.3 / 98.4 / 98.7 / 99.0	96.0 / 98.9 / 99.6 / 99.6	91.1 / 97.4 / 98.9 / 99.3
	SuperGlue	85.6 / 89.9 / 91.9 / 91.5	93.0 / 98.1 / 98.4 / 99.0	93.8 / 98.7 / 99.4 / 99.6	90.5 / 92.6 / 98.7 / 99.3

Table 1. Merged performance table (Recall@N): retrieval-only and always-on geometric re-ranking across SF_XS, TOKYO_XS, SVOX (Sun), and SVOX (Night), with best results in bold.

Global	Local	SF_XS s/q	TOKYO_XS s/q	SVOX (Sun) s/q	SVOX (Night) s/q
CosPlace	None	0.02	0.01	0.02	0.02
	LoFTR	4.68	4.26	4.27	4.30
	SP-LightGlue	4.84	4.87	4.83	4.87
	SuperGlue	1.41	1.45	1.46	1.45
MixVPR	None	0.01	0.02	0.02	0.02
	LoFTR	1.21	3.64	4.76	4.63
	SP-LightGlue	4.86	4.86	4.89	5.12
	SuperGlue	0.99	0.98	1.48	1.46
NetVLAD	None	0.04	0.04	0.04	0.07
	LoFTR	3.72	3.63	4.33	4.29
	SP-LightGlue	3.56	3.49	4.95	5.28
	SuperGlue	1.43	1.43	1.48	1.42
MegaLoc	None	0.11	0.10	0.25	0.10
	LoFTR	1.15	1.14	4.47	3.72
	SP-LightGlue	1.05	1.01	5.04	3.55
	SuperGlue	1.08	1.02	1.47	1.44

Table 2. Runtime comparison (average seconds per query) across SF_XS, TOKYO_XS, SVOX (Sun), and SVOX (Night), including retrieval-only and always-on geometric re-ranking (LoFTR, SP-LightGlue, SuperGlue).

4.2. Retrieval and Re-ranking Performance

Table 1 highlights clear and systematic differences across pipelines, driven by (i) the robustness of the global descriptor and (ii) the ability of geometric verification to resolve perceptual aliasing within the top-K shortlist. Overall localization performance depends on whether retrieval places the correct location within the shortlist and whether local matching plus RANSAC can correctly reorder visually similar candidates.

Impact of geometric re-ranking and available headroom. Geometric re-ranking yields the largest Recall@1 improvements when retrieval has headroom, i.e., when the correct place is often retrieved but not ranked first. This effect is strongest for CosPlace and NetVLAD across datasets. For example, on SF-XS CosPlace improves from 63.1 to 77.4+ Recall@1, and on Tokyo it increases from 65.1 to 84.8 with LoFTR or SuperPoint+LightGlue. NetVLAD shows similar relative gains despite a lower baseline. These patterns indicate frequent *misranking* under perceptual aliasing: global descriptors retrieve a plausible shortlist but cannot reliably order hard negatives, whereas geometric verification enforces projective consistency and promotes the true match.

Strong retrieval limits re-ranking gains. MegaLoc achieves consistently high Recall@1 across all datasets, including SVOX Night (96.5), leaving limited room for improvement. In this near-saturated regime, re-ranking provides small or negligible gains because the top-1 prediction is already correct most of the time. Moreover, small drops observed for some MegaLoc + matcher combinations can occur when matching fails (limited overlap, extreme viewpoint changes, motion blur, night noise) or when inlier scores are imperfectly calibrated across candidates. This confirms that re-ranking is most beneficial when retrieval ambiguity exists and the correct place is present but not top-ranked.

Domain shift and descriptor robustness. SVOX Night reveals the strongest contrast between retrieval models. CosPlace and especially NetVLAD degrade sharply in retrieval-only mode (33.3 and 8.5 Recall@1, respectively), reflecting limited robustness to extreme illumination changes. In comparison, MixVPR (62.6) and MegaLoc (96.5) remain substantially more robust, suggesting stronger invariance due to aggregation design, training diversity, and/or regularization. Importantly, geometric re-ranking recovers a large fraction of the lost performance for weaker descriptors, indicating

that local geometric consistency remains informative even under strong appearance shift.

Comparison of matchers: accuracy versus stability. Across retrieval backbones, LoFTR often yields the strongest Recall@1 after re-ranking in challenging scenarios (notably SVOX Night), consistent with the benefit of dense correspondences under low texture and illumination changes. However, this comes with higher computational cost and sometimes increased sensitivity to noise. SuperPoint+LightGlue is typically competitive and stable across datasets, while SuperGlue tends to produce fewer but more selective matches and often offers a favorable runtime/accuracy trade-off.

Recall@N trends and misranking analysis. The gap between Recall@1 and Recall@20 is informative. When Recall@1 is low but Recall@20 remains high (e.g., CosPlace on SVOX Night), the correct place is often retrieved but misranked. This is exactly the regime where geometric re-ranking is most effective: it corrects ordering errors inside the shortlist rather than compensating for retrieval failures. Conversely, when Recall@20 is already near saturation (as for MegaLoc), re-ranking has limited upside.

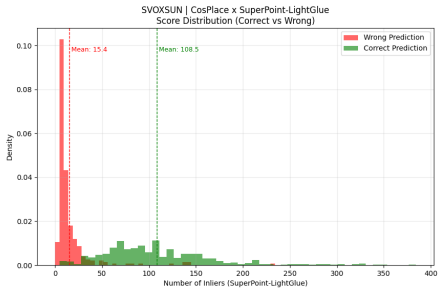
4.3. Runtime and Cost Analysis

Table 2 shows that computation is dominated by geometric re-ranking. Retrieval-only pipelines are extremely fast (0.01–0.11 s/query), whereas always-on verification costs multiple seconds per query when matching all $K = 20$ candidates.

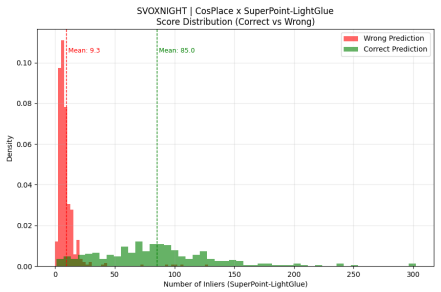
Cost scaling with K. In the standard pipeline, the matcher and RANSAC are executed independently for each of the top-K candidates, yielding an overall cost of approximately $t_r + Kt_m$. Even modest per-match runtime becomes prohibitive when multiplied by K, which limits scalability for real-time or large-scale deployment.

Matcher-dependent cost differences. SuperGlue is consistently the fastest matcher (often $\sim$1.4 s/query), while LoFTR and SuperPoint+LightGlue are typically slower (often $\sim$4–5 s/query). These differences are consistent with architectural choices: LoFTR performs dense transformer-style matching, SuperPoint+LightGlue adds detection and description stages, while SuperGlue operates on a bounded number of keypoints.

Dataset influence on runtime. Runtime varies across datasets because texture richness, overlap, and scene complexity affect the number of valid correspondences and RANSAC behavior (including iteration count and early stopping). More challenging conditions (e.g., night) can increase matching difficulty and thus runtime.



(a) SVOX Sun: CosPlace $\times$ SuperPoint-LightGlue.



(b) SVOX Night: CosPlace $\times$ SuperPoint-LightGlue.

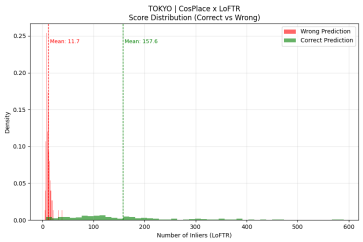
Figure 2. Distribution of top-1 inlier scores I_1 split by whether the retrieval top-1 prediction is correct or wrong. A strong shift toward higher inliers for correct predictions supports using I_1 as a confidence signal for adaptive re-ranking, while the domain shift suggests using a domain-aware gate.

Implication: motivation for selective verification. While geometric re-ranking improves accuracy, its cost motivates compute-aware strategies. If many queries are easy, paying for K matchings every time is inefficient, which directly motivates the adaptive extension.

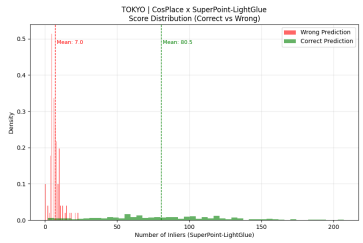
4.4. Inlier-score distributions and implications for adaptive gating

To support the adaptive policy, we analyze the distribution of the top-1 geometric score (I_1, number of RANSAC inliers) for *correct* vs *wrong* top-1 retrieval predictions. Figures 2 and 3 visualize these distributions for representative pipelines.

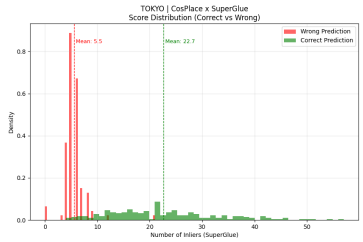
On SVOX, CosPlace $+$ SuperPoint-LightGlue shows a clear separation: inliers for wrong predictions concentrate at very small values, while correct predictions shift to much larger inlier counts. This effect holds in both Sun and Night, but the absolute inlier scale changes, motivating domain-specific calibration (LR-Sun vs LR-Night). On Tokyo, the inlier scale and separation depend on the matcher: LoFTR produces larger raw inlier counts, while SuperGlue produces lower counts but still separates correct vs wrong predictions. This motivates learning a calibrated gate rather than using a single fixed threshold.



(a) Tokyo: CosPlace × LoFTR.



(b) Tokyo: CosPlace × SuperPoint-LightGlue.



(c) Tokyo: CosPlace × SuperGlue.

Figure 3. Tokyo inlier-score distributions for three matchers. The raw inlier scale depends on the matcher, motivating calibration (LR gating) rather than a single fixed threshold across pipelines.

5. Adaptive Re-ranking Extension: Results and Analysis

We report adaptive re-ranking results on Tokyo_XS, SVOX (Sun/Night), and SF_XS. SF_XS is generally easier than SVOX Night and is included to show how the gate behaves in a less ambiguous regime. For all experiments, $K = 20$ and the positive distance threshold is 25 m. We report adaptive Recall@1, and compute proxies: hard rate (fraction triggering full re-ranking) and savings vs. full $K = 20$ matching; in practice, these computational savings generally translate into reduced runtime.

Tokyo_XS. On Tokyo_XS, adaptive re-ranking consistently improves Recall@1 while evaluating far fewer than K matchings on average. For MixVPR-based pipelines, Recall@1 typically increases from approximately 76–77 to 85–88, corresponding to compute savings of roughly 70–80%. This indicates that many Tokyo queries are moderately ambiguous: retrieval alone often misranks visually similar negatives, but only a small number of geometric probes is sufficient to decide whether full verification is required. CosPlace-based pipelines exhibit even larger absolute gains (often +15 to +18 Recall@1 points), which is consistent with their weaker retrieval robustness and therefore larger headroom for geometric re-ranking. Overall, Tokyo XS highlights the effectiveness of adaptive gating in correcting retrieval errors at a limited computational cost.

SVOX Sun. SVOX Sun is particularly favorable to adaptive gating. For strong retrieval backbones such as MixVPR, hard-query rates remain low, typically in the 10–15% range, resulting in compute savings exceeding 80%. Despite this aggressive pruning, adaptive Recall@1 still improves by several points compared to retrieval-only baselines. This behavior suggests that under daylight conditions, the majority of queries are "easy": the top-ranked retrieval is already reliable, and geometric verification is only required for a small fraction of ambiguous cases. As a result, adaptive re-ranking achieves an excellent accuracy–compute trade-off in this regime.

SVOX Night. SVOX Night remains the most challenging evaluation setting. Retrieval-only Recall@1 is substantially lower, especially for CosPlace, which creates significant headroom for re-ranking. Adaptive re-ranking recovers large accuracy gains, often on the order of +20 Recall@1 points for CosPlace and +15/+18 points for MixVPR, but at the cost of higher hard-query rates, frequently exceeding 35–60%. This behavior is expected: severe illumination changes reduce the reliability of the top-1 retrieval, causing the gate to trigger full re-ranking more often. Importantly, even in this difficult regime, adaptive re-ranking still avoids paying the full $K = 20$ matching cost for every query, while preserving most of the accuracy gains of always-on geometric verification.

SF_XS. SF_XS is comparatively easier, which is reflected in consistently low hard-query rates across methods. Adaptive re-ranking reliably improves Recall@1 while maintaining very high compute savings. LoFTR-based variants typically achieve savings on the order of 80–85% with moderate but consistent accuracy gains, whereas SuperPoint+LightGlue variants tend to provide larger Recall@1 improvements at the cost of higher hard rates and reduced savings. This dataset clearly illustrates the accuracy–compute trade-off enabled by adaptive gating, which can be tuned by the choice of local matcher and gating strategy.

5.0.1. Threshold sensitivity and cross-dataset calibration

Thresholds tuned on SF_XS may be suboptimal on other datasets (SVOX Sun/Night) due to different ambiguity and inlier statistics. Still, Recall@1 increases rapidly at low thresholds and then saturates across datasets, revealing a

Dataset	Method (LR gate)	Adapt R@{1/5/10/20}	Hard (%)	Savings (%)
Tokyo	MixVPR + LoFTR (LR-Sun)	84.8/91.1/92.7/94.2	**16.5**	**79.3**
	MixVPR + LoFTR (LR-Night)	87.0/92.7/93.6/94.2	20.3	75.7
	MixVPR + SP-LightGlue (LR-Sun)	**88.3**/92.7/93.6/94.2	25.1	71.2
	MixVPR + SP-LightGlue (LR-Night)	87.9/92.7/93.6/94.2	26.0	70.3
	CosPlace + LoFTR (LR-Sun)	79.7/85.7/88.3/89.5	28.9	67.6
	CosPlace + LoFTR (LR-Night)	82.2/85.7/87.9/89.5	33.3	63.3
	CosPlace + SP-LightGlue (LR-Sun)	82.9/86.7/88.2/89.5	37.8	59.1
	CosPlace + SP-LightGlue (LR-Night)	82.9/87.0/88.6/89.5	39.0	57.9
SVOX Sun	MixVPR + LoFTR (LR-Sun)	88.3/94.0/95.1/95.4	**10.5**	**85.0**
	MixVPR + LoFTR (LR-Night)	**89.1**/94.1/95.2/95.4	12.8	82.9
	MixVPR + SP-LightGlue (LR-Sun)	88.9/94.0/95.0/95.4	13.3	82.3
	MixVPR + SP-LightGlue (LR-Night)	88.8/93.9/95.0/95.4	14.4	81.3
	CosPlace + LoFTR (LR-Sun)	72.0/83.2/86.1/88.7	20.8	75.2
	CosPlace + LoFTR (LR-Night)	75.5/84.4/87.0/88.7	25.5	70.7
	CosPlace + SP-LightGlue (LR-Sun)	76.1/84.5/87.8/88.7	27.4	69.0
	CosPlace + SP-LightGlue (LR-Night)	77.2/85.1/87.8/88.7	30.1	66.4
SVOX Night	MixVPR + LoFTR (LR-Sun)	78.4/84.9/86.5/87.4	**33.7**	**63.0**
	MixVPR + LoFTR (LR-Night)	80.1/85.3/86.5/87.4	36.9	59.9
	MixVPR + SP-LightGlue (LR-Sun)	79.8/85.3/86.3/87.4	35.2	61.5
	MixVPR + SP-LightGlue (LR-Night)	**80.9**/85.3/86.3/87.4	37.4	59.4
	CosPlace + LoFTR (LR-Sun)	54.3/62.2/63.5/67.6	56.0	41.8
	CosPlace + LoFTR (LR-Night)	56.4/62.8/64.0/67.6	60.9	37.2
	CosPlace + SP-LightGlue (LR-Sun)	57.0/63.2/65.0/67.6	60.9	37.2
	CosPlace + SP-LightGlue (LR-Night)	58.3/63.8/65.2/67.6	63.4	34.7
SF-XS	MixVPR + LoFTR (LR-Sun)	71.8/79.6/82.0/83.6	**11.1**	**84.5**
	MixVPR + LoFTR (LR-Night)	72.6/80.0/82.2/83.6	14.2	81.5
	MixVPR + SP-LightGlue (LR-Sun)	77.8/81.6/83.0/83.6	24.8	71.4
	MixVPR + SP-LightGlue (LR-Night)	**78.4**/81.7/83.1/83.6	26.3	70.0
	CosPlace + LoFTR (LR-Sun)	66.5/76.0/79.4/81.4	11.5	84.1
	CosPlace + LoFTR (LR-Night)	67.5/76.4/79.5/81.4	15.7	80.1
	CosPlace + SP-LightGlue (LR-Sun)	73.8/79.1/80.7/81.4	29.1	67.4
	CosPlace + SP-LightGlue (LR-Night)	74.5/79.2/80.7/81.4	30.7	65.8

Table 3. Adaptive LR-gated matching results: adaptive Recall@N, hard rate, and savings vs. full K=20 matching. Best Adapt R@1 and best efficiency per dataset are bolded.

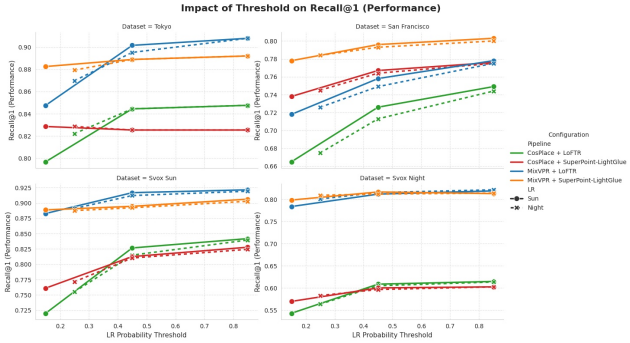


Figure 4. Impact of the logistic-regression probability threshold τ on adaptive Recall@1 across datasets and pipelines.

stable "knee" region and robust generalization under domain shift.

6. Conclusion

We presented a two-stage VPR pipeline combining global descriptor retrieval with local geometric re-ranking. We discussed how major retrieval models differ in aggregation and robustness, and how modern matchers with RANSAC provide a geometric verification signal via inlier counting. Our experiments show that re-ranking can substantially improve Recall@1 in challenging settings (notably night domain shift and weaker retrieval baselines), but at a high computational cost. We also confirmed that dot product and ℓ_2 retrieval yield identical rankings due to descriptor normalization.

Finally, we introduced an adaptive re-ranking extension that reduces average computation by verifying only hard queries. The method uses a top-1 inlier probe and a logistic regression gate, with domain-specific LR-Sun/LR-Night models, and is evaluated with runtime-aware metrics (hard rate and savings vs. full $K = 20$ re-ranking). Overall, adaptive gating offers a practical and principled accuracy–compute trade-off for VPR systems.

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